



Ropener Curtain Controller

Bill of Materials

Version 2.0

Everything needed to build one Ropener. Quantities are per unit. Links in the Description column point to the custom PCB and suggested suppliers; equivalent parts may be substituted where noted.

Body Assembly

Part	Qty	Description
VAL3000 PCB	1	Custom PCB designed for this project
NEMA 17 motor, 48 mm body	1	Smaller length NEMA 17 will also work
12V / 2A+ power adapter w/ 5.5×2.1 mm plug	1	Technically 5–29 V will work, but 12 V is best due to low speeds
Extension cable	1	Extends the length of the power adapter
M3 × 10 mm screw, self-tapping	8	
M3 × 35 mm screw, button head	3	
M3 square nut	1	
M3 × 16 mm screw, button head	2	
M3 × 10 mm screw, flat head	2	
Gear	1	MK7
Bearing	2	7 mm × 3 mm × 3 mm

Curtain Assembly

Part	Qty	Description
PTFE tubing, 3 mm ID × 5 mm OD	1	290 mm length required
M3 × 10 mm screw, self-tapping	11	
M3 square nut	1	
M3 × 12 mm screw, flat head	1	
Zip tie	1	Small, 2–3 mm wide
V623ZZ pulley	1	3 × 12 × 4 mm size
Rope	1	1.6 mm diameter

Mounting Hardware

Part	Qty	Description
Command strips	2	The easiest way to mount to any wall
#17 × 1 1/4" wire nail	3	Use to nail to drywall only

Assembly Tools Required

Part	Qty	Description
Razor	1	To cut the length of PTFE tubing
Screwdriver, Phillips No. 1	1	
Hex driver, 2 mm	1	